

Customer Churn Prediction – Summary Report (Extra Section)

1. Problem statement

Predict which subscription customers are likely to churn, identify key drivers of churn, and support targeted retention strategies.

2. Methodology (extra analysis)

- Used the existing churn dataset to run additional EDA on contract type, internet service and senior citizen segments.
- Built a compact modeling pipeline based on scikit-learn with a standardized numeric + one-hot encoded categorical feature space.
- Trained two supervised models: Logistic Regression and Random Forest, and evaluated them on a held-out test set.

3. Extra best model performance

- Best model: Logistic Regression (extra)
- Accuracy: 0.804
- Precision: 0.648
- Recall: 0.575
- F1: 0.609
- Roc_auc: 0.836

4. Example insights (to be refined based on EDA)

- Month-to-month contracts and short-tenure customers typically exhibit higher churn rates.
- Certain internet service types and higher monthly charges may be associated with higher churn risk.
- Senior citizens can have different churn behaviour that justifies tailored communication and offers.

5. Recommendations (summary)

- Use the best performing model to score active customers regularly and focus retention campaigns on the highest risk group.
- Prioritise high-value customers in the high-risk group for proactive outreach and tailored offers.
- Review onboarding and perceived value for short-tenure customers and those on flexible contracts.
- Consider further model tuning and adding behavioural features (usage, support tickets, marketing interactions).

End of extra report.